

REPRODUCIBLE BENCHMARK

PostgreSQL vs DuckDB on the same data

A seeded, verified, open benchmark of analytical query execution time — not HTTP latency, not cached responses.

12 queries

30 iterations each

100K + 1M rows

seed 42

results verified equal

METHOD

What is actually measured

- Database execution time, timed at the driver call
- No HTTP, no Express, no auth, no JSON, **no app cache**
- Both engines keep **all production indexes**; PostgreSQL is ANALYZEd
- Results compared row-by-row **before** any timing is recorded
- Execution order alternates so neither engine goes first every time

PostgreSQL 16.15

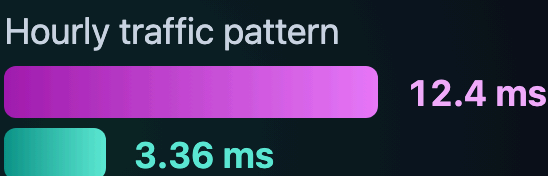
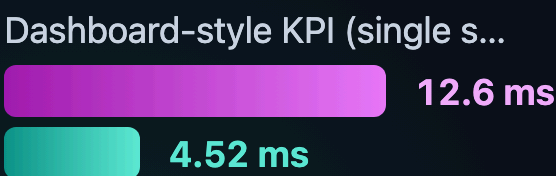
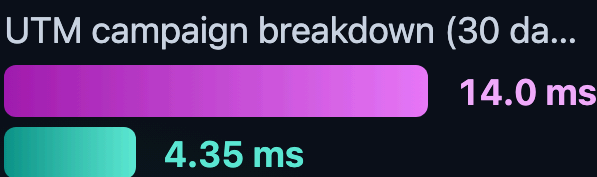
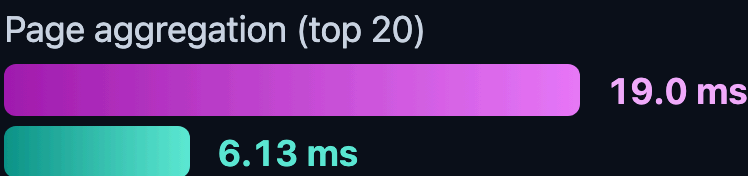
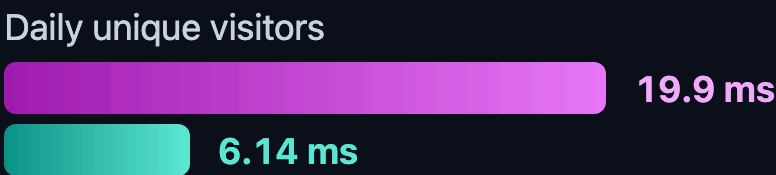
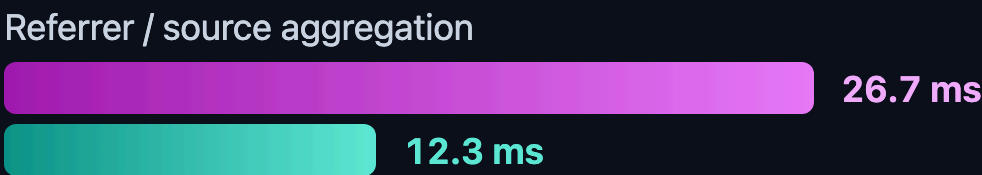
DuckDB v1.4.4

Node v22.19.0

5 warmup + 30 measured

Slowest 8 queries

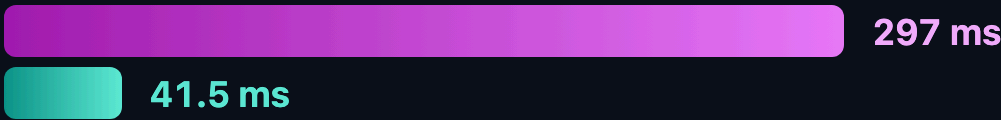
■ PostgreSQL ■ DuckDB



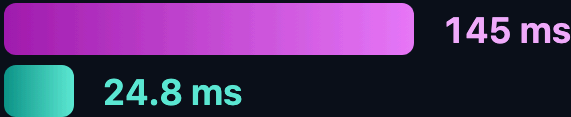
Slowest 8 queries

■ PostgreSQL ■ DuckDB

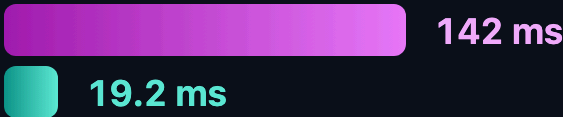
Referrer / source aggregation



Page aggregation (top 20)



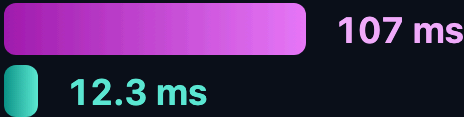
Daily unique visitors



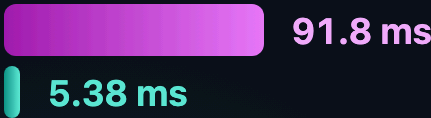
UTM campaign breakdown (30 da...



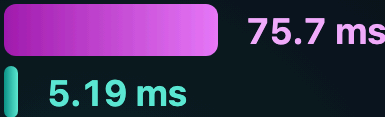
Dashboard-style KPI (single s...



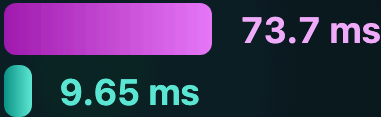
Hourly traffic pattern



Daily event aggregation



Date-range filter (30 days, s...



1M EVENTS • ALL 12 QUERIES

Median & p95 (ms)

QUERY	PG MED	DUCK MED	PG P95	DUCK P95
Total events	9.19	0.88	9.82	1.29
Unique visitors	36.8	7.55	45.8	16.4
Total sessions	42.8	8.71	43.7	15.3
Daily event aggregation	75.7	5.19	78.9	6.96
Daily unique visitors	142	19.2	146	27.4
Page aggregation (top 20)	145	24.8	169	40.1
Referrer / source aggregation	297	41.5	320	65.0
Date-range filter (30 days, singl...	73.7	9.65	78.5	15.4
Multi-dimensional GROUP BY	58.7	6.82	89.7	16.4
Dashboard-style KPI (single site,...	107	12.3	122	24.1
UTM campaign breakdown (30 days)	118	8.97	133	19.1
Hourly traffic pattern	91.8	5.38	95.5	9.10

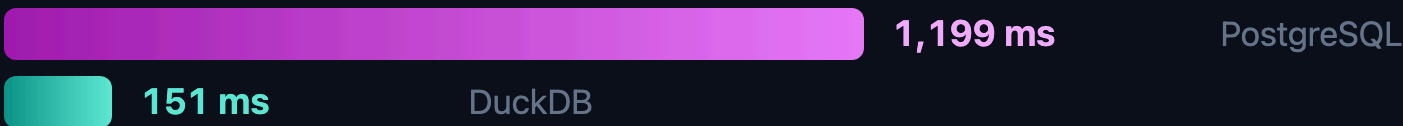
30 iterations each

full distribution in JSON

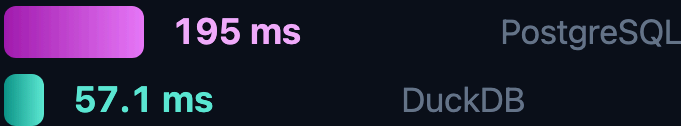
OVERALL PERFORMANCE

Total time for all 12 queries

1,000,000 events



100,000 events

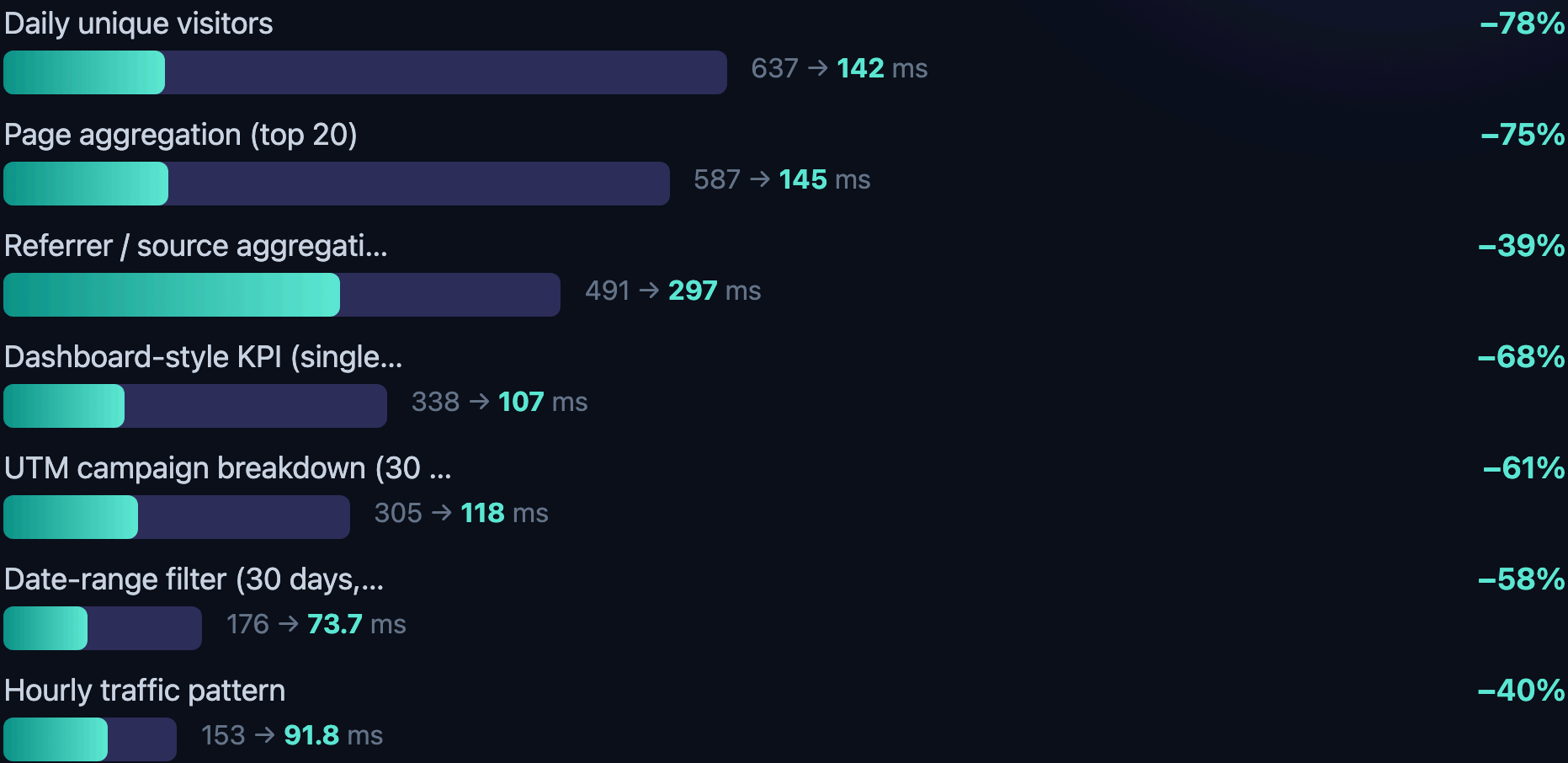


Sum of per-query medians, 30 measured iterations each. Database execution time only.

1M EVENTS · POSTGRESQL

Three rounds of tuning

Every change driven by EXPLAIN (ANALYZE, BUFFERS) — planner cost model, then query form.



Whole workload **2,972 ms** → **1,199 ms** (−60%)

READ THIS BEFORE QUOTING IT

What this does **not** say

- Not a universal claim — one workload, one machine, one config
- PostgreSQL was **tuned in its favour** after profiling, not left on defaults
- Warmed cache, not cold-start
- Reads only — ingest was **not** measured (DuckDB loaded slower here)
- Ratios vary $\sim 5\times$ – $17\times$ by query; a single "Nx faster" would mislead

Run it yourself:

```
npm run benchmark:engine -- --size 1000000 --seed 42
```